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Nanoengineering Heterogeneous Catalysts by Atomic Layer Deposition

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Abstract

A new generation of catalysts is needed to meet society's energy and resource requirements. Current catalyst synthesis does not fully achieve optimum control of composition, size, and structure. Atomic layer deposition (ALD) is an emerging technique that allows for synthesis of highly controlled catalysts in the form of films, nanoparticles, and single sites. The addition of ALD coatings can also be used to introduce promoters and improve the stability of traditional catalysts. Evolving research shows promise for applying ALD to understand catalytically active sites and create next-generation catalysts using advanced 3D nanostructures.

INTRODUCTION

The planet's rapidly growing population is projected to reach 10 billion people by the end of this century, resulting in an ever-increasing need for a sustainable supply of energy, food, and commodity chemicals. To relieve the heavy dependence on petroleum sources and reduce greenhouse gas emissions in these key areas, renewable carbon sources, such as biomass and CO₂, and renewable energy sources, such as solar, need to be coupled to meet society's future needs in a sustainable and environmentally benign manner. Efficient chemical transformations from renewable resources to commodity chemicals and fuels are highly desired. Heterogeneous catalysis plays a crucial role in tackling this grand challenge. Currently heterogeneous catalysis is used in approximately 90% of all chemical production and 20% of industrial products (1). However, gaining a deep understanding of the catalytic properties, improving existing catalysts, and rationally designing a new generation of catalysts are required to achieve lower energy intensity, better selectivity, higher activity, and the ability to catalyze new classes of reactions.

Many desirable reactions, such as the conversion of biomass to fuels and chemicals (2), water splitting (3), CO hydrogenation to higher alcohols (4), partial methane oxidation (4), and the conversion of CO₂ to fuels and chemicals (4), currently are not industrially feasible owing to inadequate catalysts. This motivates active catalyst development as well as fundamental research into catalyst reactivity and selectivity. Although highly controlled, monodisperse catalysts are desirable, traditional synthetic methods for heterogeneous catalysts provide limited control (5) and lead to catalysts with mixtures of sizes and composition. This variability can result in the presence of different types of surface sites, leading to both wasted active material and potentially unwanted side reactions. Hence, in the search for high-performance catalysts, there is a need for control over the catalyst size, composition, and structure. Further, there is demand for new catalyst synthesis methods with atomic-level control to create structured catalysts with precise active sites that can achieve a level of activity and selectivity not afforded by traditional synthetic methods.

Atomic layer deposition (ALD) is an emerging materials synthesis technique that may be able to meet the needs for the design and synthesis of heterogeneous catalysts. ALD allows for building catalytic surfaces, creating active phases, and improving stability in a highly controlled and uniform manner (6–11). In this review, we cover the fundamentals of ALD with a focus on its application for heterogeneous catalysis. Heterogeneous catalysts typically consist of nanoparticles (e.g., metals, alloys, metal oxides, metal sulfides, or metal carbides) on a supporting material, often with additives and promoters, and ALD can be applied to synthesize or modify almost all the component parts of a heterogeneous catalyst, as described below.

ATOMIC LAYER DEPOSITION

Brief Overview of Atomic Layer Deposition

ALD is a thin-film deposition technique based on sequential, self-limiting surface reactions between vapor-phase precursors and a solid substrate surface (12, 13). Compared with other film deposition techniques, including chemical vapor deposition (CVD) and physical vapor deposition (PVD), the characteristic self-limiting growth mechanism of ALD enables excellent film conformality; pinhole-free film deposition; accurate thickness control at the angstrom level; tunable film composition; surface chemical selectivity; and penetration into high-aspect-ratio, porous, and 3D substrates (14–19).

Although initially introduced in the 1960s and 1970s for applications in flat-panel displays (20, 21), during the past decades ALD has expanded greatly to include many industrial and research processes (22–26). The rapidly growing microelectronics industry has driven the fast

development of ALD owing to a desire for high-quality thin-film deposition on devices with shrinking dimensions and complex patterns (27, 28). In recent years, the application of ALD has expanded to benefit materials fabrication in other fields, including photovoltaics (24, 29) and energy storage (25). Regarding catalysis, initial work performed in the late 1990s demonstrated ALD as a means for preparing heterogeneous catalysts with better control over the catalyst properties compared with conventional methods (19, 30–33). Recently, there has been a revival of ALD in heterogeneous catalysis applications (6, 10, 11).

The advantages of ALD stem from the self-limiting nature of the gas-surface reaction. Unlike CVD, where all the reagents for film synthesis are introduced simultaneously, ALD exposes the substrate to the two reagents separately as two sequential half-reactions. The two reagents involved in the ALD process are the precursor (inorganic or metalorganic) and the counter-reactant (e.g., H_2O , O_2 , O_3 , H_2). **Figure 1** illustrates one typical ALD cycle, which is usually conducted in a vacuum chamber. The precursor vapor is first pulsed into the chamber, where it reacts with surface species (e.g., hydroxyl groups) (34, 35) or adsorbs on energetic defect sites (36, 37). Such half-reactions are limited by the number of available surface adsorption sites, constraining the adsorption of the precursor to one monolayer or, more typically, submonolayer. Inert gas flow and evacuation remove residual precursor molecules and by-products. The second half-reaction takes place in a similar manner by introducing the counter-reactant that reacts with the functional groups on the precursor-saturated surface and thus completes one ALD cycle. The film thickness can be controlled by varying the number of repeated ALD cycles. Given sufficient time and space for diffusion, the gaseous reactants are able to penetrate into porous structures, deep trenches, and complex 3D geometries, allowing for conformal coatings, whereas CVD and PVD often suffer from nonuniformity and shadowing effects when applied to high-aspect-ratio structures (19, 38). The uniform and controllable ALD oxide coatings achieved on anodic aluminum oxide (39) and mesoporous SiO_2 (40, 41) shown in **Figure 1** clearly demonstrate this advantage.

Atomic Layer Deposition for Heterogeneous Catalysis

The application of ALD in heterogeneous catalysis has rapidly gained interest. This trend is not surprising, given the attributes of ALD that allow for controllable materials synthesis even in porous substrates and may provide some advantages over conventional catalyst synthesis techniques. Heterogeneous catalyst substrates usually have high surface area or a porous structure to maximize reactivity. Therefore, the capability of uniform coating on these substrates enabled by ALD is highly valuable.

Conventional ALD is conducted inside a vacuum chamber where the sample substrate (a planar or otherwise monolithic sample) statically resides. To ensure effective precursor penetration and coating on powder or highly porous samples, modifications of both the ALD process and reactor design have been developed. Lengthened pulse and purge times are typically adopted to allow for full penetration of the ALD precursors, as well as isolation of the deposition chamber from the vacuum pump when precursors are dosed to maximize exposure for a given amount of precursor. Particles can be coated in a conventional ALD system by containing the particles within a metal dish (42). Specialized systems based on fluidized bed reactors (43), rotary reactors (44), and spatial reactors (45) have all been developed to coat particles. For more details about the design of ALD reactors for catalysis applications, the reader is referred to a prior review (46). Among the commonly used catalyst supports, mesoporous silica (47, 48), powders of nanometer size (49–51), zeolites (52–55), nanotubes (56), nanowires (57, 58), aerogels (59, 60), anodic aluminum oxide (61), and metal organic frameworks (MOFs) (62) have all been uniformly coated by ALD. Some of these examples are shown in **Figure 2**.

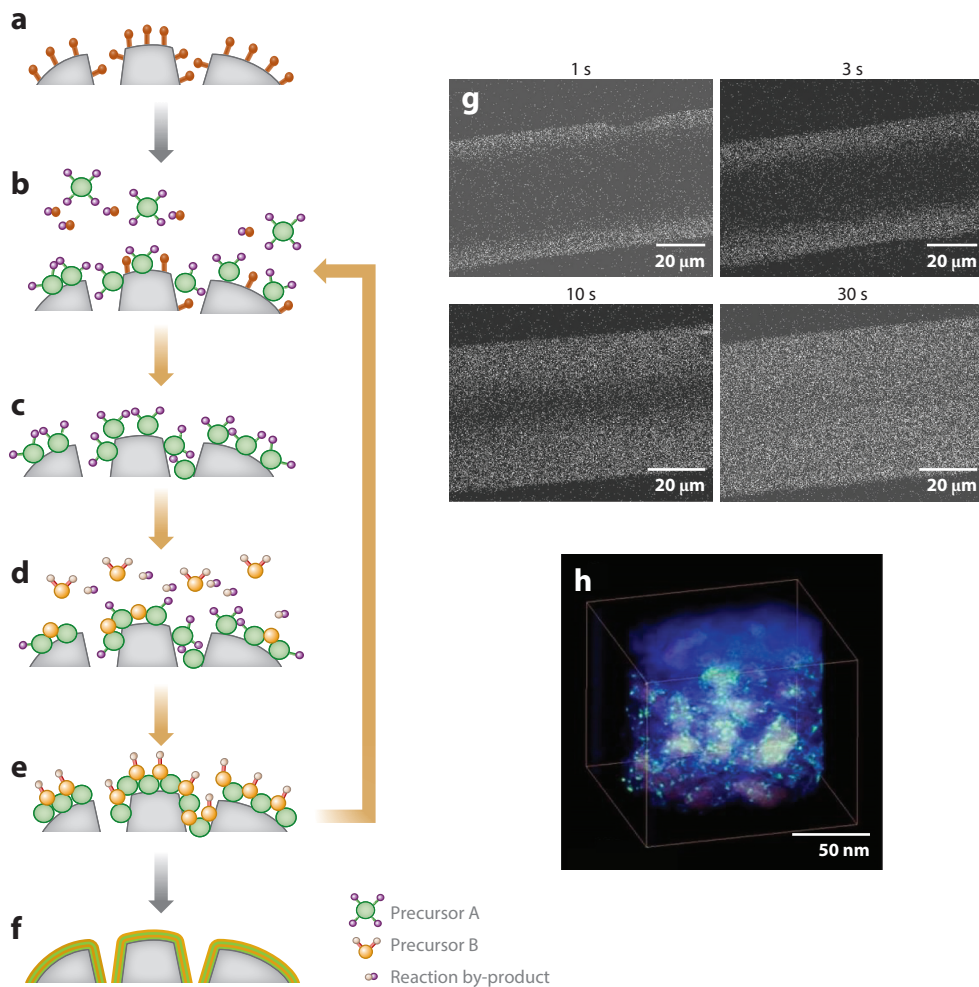


Figure 1

Schematic of atomic layer deposition (ALD) process. (a) Substrate surface with functional groups or highly energetic anchoring sites. (b) Precursor A is introduced and reacts with the surface. (c) Precursor A has reached saturation; excess precursor and by-products have been purged by inert gas. (d) Precursor B is introduced and reacts with the surface. (e) Precursor B has reached saturation; one (sub)monolayer of material is formed. (f) After repeating steps b–e for a set number of cycles, a film with the desired thickness is obtained on the substrate. (g) Electron probe microanalysis map of 64 cycles of ZnO ALD on anodic aluminum oxide, using various precursor exposure times. White dots represent Zn and longer exposure improves the uniformity of ZnO deposition inside the membrane. Reprinted with permission from Reference 39. Copyright 2003, Am. Chem. Soc. (h) Three-dimensional tomographic reconstruction of mesoporous silica with 50 cycles of TiO₂ ALD with mesopores in white and anatase titania in green, showing titania deposition throughout the mesoporous structure. Reproduced from Reference 41 with permission of R. Soc. Chem.

In addition to the capability to produce uniform coatings on high-surface-area substrates, a wide range of materials have been grown by ALD, including oxides, sulfides, nitrides, and pure elements (15). These ALD processes are readily available for the synthesis of many typical catalyst supports, promoters, and active phases, with many oxides (e.g., SiO₂, TiO₂, Al₂O₃, ZnO, ZrO₂, CeO₂, La₂O₃, Fe₂O₃, MgO, MnO), and metal phases successfully prepared (e.g., Pt, Pd, Ru, Rh, Co, Ni). Although ALD is a thin-film deposition technique, different growth regimes exist,

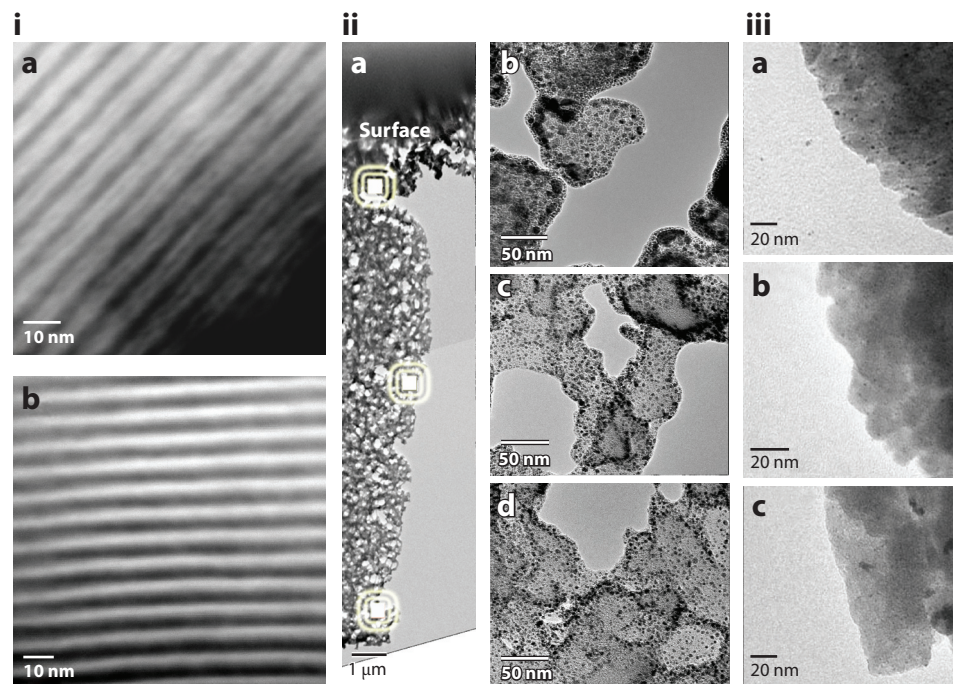


Figure 2

Atomic layer deposition (ALD) coating of various porous catalyst substrates: (i) high-magnification scanning transmission electron microscopy image of mesoporous silica (type SBA-15) after (a) and before (b) 30 cycles of niobia ALD. In *i*–*b*, the bright lines represent silica walls, whereas the dark lines represent empty pore space. After niobia deposition, which appears brighter than silica, in *i*–*a*, the thin dark lines sandwiched within bright lines represent the silica walls. In *i*–*b*, the lack of such contrast across the bright line indicates that the silica walls are not coated with niobia. Reprinted with permission from Reference 47, ©2011, Am. Chem. Soc. (ii) Cross-sectional transmission electron microscopy images of a carbon aerogel with Pt nanoparticles deposited by ALD. Subpanels *b*–*d* were taken at 1, 5, and 10 μm beneath the surface, corresponding to the marked positions in subpanel *a*, showing the penetration of Pt ALD into the interior surface of the carbon aerogel. Reprinted with permission from Reference 60, ©2008, Am. Chem. Soc. (iii) Ir on beta-zeolite: (a) deposited by ALD using $\text{Ir}(\text{acac})_3$ as precursor; (b) with 2,4-pentanedione as support surface treatment prior to Ir ALD; (c) deposited by impregnation. Ir prepared by ALD shows better uniformity in metal size distribution. Support pretreatment with 2,4-pentanedione decreases Ir particle size. Reprinted from Reference 53 with permission from Springer.

starting from the initial period in which continuous films, islands, or nanoparticles can be formed, depending on the specific ALD chemistry (34). For example, the deposition of noble metals on oxide surfaces can experience a nucleation delay owing to slow nucleation kinetics; this combined with a tendency for metal nanoparticles to agglomerate (63) can lead to deposition of nanoparticles. In the field of heterogeneous catalysis, both films and nanoparticles are highly desired, depending on the application. On the one hand, ALD of continuous films provides high-quality, uniform coatings with accurate thickness control that can be applied to finely tune the surface properties of a catalyst support without affecting its bulk properties. Moreover, films deposited as overcoating layers can enhance stability, introduce new active sites, or tune selectivity. On the other hand, ALD can also deposit nanoparticles composed of metals, alloys, oxides, and sulfides useful for the active catalyst material.

Some potential benefits of ALD nanoparticle synthesis compared with conventional catalyst synthesis that is typically conducted in solution (10, 64)—such as impregnation, precipitation, and colloidal synthesis—include monodisperse size, uniform composition, lower contamination from residual salts or capping agents, and the ability to penetrate highly porous supports. Other gas-phase deposition techniques, such as PVD, have been demonstrated for catalyst synthesis as well (65); however, unlike ALD, these methods are not as well optimized for uniformly coating the interior of highly porous catalyst substrates. By optimizing ALD conditions (such as growth temperature, choice of precursor and counter-reactant, and precursor pulse length) and designing novel ALD strategies [e.g., area-selective deposition (66) or manipulation of pulsing sequences (67)], desirable catalyst nanostructures and surface properties can be achieved. Such fine tuning also provides a means for gaining a fundamental understanding of the active sites, reaction pathways, and degradation mechanisms of a heterogeneous catalyst system.

Several excellent reviews have been written about ALD in general (12–15, 18, 23, 34) and the application of ALD to catalysis in particular (6, 8, 11, 68, 69). The focus of this review is on recent examples that illustrate how ALD has improved the synthesis and understanding of heterogeneous catalysts. For more comprehensive summaries of the process, the reader is referred to the earlier reviews.

APPLICATIONS OF ATOMIC LAYER DEPOSITION TO HETEROGENEOUS CATALYSIS

Synthesis of the Active Catalyst

There are several motifs in which ALD can be used to create the active sites on a catalyst: as a film, as individual sites, and as nanoparticles.

Catalytically active ALD films have been applied to a variety of supports. Deng et al. (70) investigated the use of cobalt ALD coatings on nanoporous anodized aluminum oxide membranes and compared them with traditionally synthesized Co clusters for the cleavage of C–O–C bonds for biofuel production. However, the authors found that the catalysts prepared using ALD had a lower activity than those prepared using highly dispersed nanoparticles, owing to the limited number of undercoordinated surface atoms on the smooth ALD films. Ivanova et al. (71) investigated CeO₂ films deposited using ALD on stainless steel foil as a diesel soot combustion catalyst. The activity of the resulting catalysts could be modulated based on the ALD deposition temperature as higher temperatures increased the concentration of the highly active Ce³⁺ species (71). On nanoporous aluminum oxide, vanadium oxide ALD was used to create cyclohexane dehydrogenation catalysts (72). The prepared catalysts demonstrated higher activity than incipient wetness impregnation (IWI)-synthesized catalysts owing to their greater dispersion (72). Keränen et al. (73) noted a similar effect for ALD-synthesized vanadia catalysts. We note that thin films grown by ALD have also found key application in the area of electrocatalysis, but such electrocatalytic systems are beyond the scope of this review, and the interested reader is referred to several references on the subject (6, 25, 74–76).

On zeolites and MOFs, ALD has been used to create catalytically active single sites through modification of the framework. Work on zeolites demonstrated that aluminum oxide ALD could be used as a means of increasing activity (54). In that study, aluminum oxide was found to incorporate into the silica matrix and increase the number of acidic catalytic sites within the zeolite framework. Similar work by Jiang et al. (52) demonstrated ZnO ALD as a means of tuning the acidity and activity of zeolite-based catalysts. Mesoporous silica-based catalysts have also been prepared through ALD. SBA-15 mesoporous silica was coated using niobia ALD to create

2-butanol and 2-propanol dehydration catalysts (47). The presence of niobia stabilizes the SBA-15 support by consuming silanol groups, resulting in a catalyst with higher hydrothermal stability than a commercial niobia catalyst (47).

Another key application is for the modification of MOFs, where ALD films can selectively deposit at metal active sites and modify the structure in a highly controlled manner. In one study, trimethyl indium and trimethyl aluminum were used to deposit on –OH groups at Zr-based MOF nodes; however, the high reactivity of trimethyl aluminum resulted in damage to the MOF structure (77). In another example, Ni grown using bis(N,N'-di-tert-butyl-acetamidinato)nickel(II) and H₂O selectively deposited on –OH groups of Zr-based MOF nodes (78). The resulting catalysts had highly dispersed Ni species that provided significant gas-phase alkene hydrogenation activity. Moreover, the catalysts prepared through Ni ALD on MOFs had substantially higher stability than conventional Ni/ZrO₂ catalysts that suffer from sintering during hydrogenation catalysis. In the Ni ALD-modified MOF, Ni sites are spatially isolated from each other in the MOF framework, inhibiting sintering of Ni.

As described in the section titled Atomic Layer Deposition for Heterogeneous Catalysis, ALD can be used to synthesize metal nanoparticles as well as thin films. The synthesis of nanoparticles using ALD is clean in the sense that precursor ligands leave as volatile species and no post-treatment is typically required to remove ligands or other impurities after synthesis (79). ALD is also a direct synthesis technique wherein particles are deposited onto the support without a separate grafting step (80). The production of nanoparticles by ALD typically originates from the specific growth mode. There are three major growth regimes for a nucleating film: thin-film growth, island growth, and thin-film-plus-island growth (38). In metal deposition, often owing to more favorable interaction energetics between the depositing metal compared with the interaction energetics between the substrate and the metal, ALD growth can proceed via an island growth mode (81, 82). Such deposition can result in the formation of well-dispersed nanoparticles. The coverage can be controlled based upon the growth conditions as well as by the use of surface modifiers, such as self-assembled monolayers (SAMs) (82). Notably, SAMs control the nucleation of the film and potentially could be used to enforce an island growth mechanism and grow nanoparticles of materials that typically deposit as thin films during ALD (81–83).

Such nucleation control schemes are not required in the case of Pt and Pd nanoparticles grown using ALD. ALD of Pt and Pd nanoparticles has found comparatively widespread use for preparing catalysts. The addition of Pt nanoparticles to a high-surface-area structure, such as for use as a cocatalyst, is difficult to perform in a uniform manner using PVD, impregnation, or other traditional deposition methods (84). This lack of uniformity lowers the catalyst's overall performance and results in wasted material where excess Pt is deposited (84). ALD therefore has some advantages for depositing Pt nanoparticles and has been applied to a variety of reactions (49, 60, 84–89). King et al. (60) used ALD to synthesize Pt nanoparticles on carbon aerogels for CO oxidation. The ALD process penetrated to a depth greater than 10 μm into the carbon aerogel, and the resulting catalysts were highly active for CO oxidation. Work by Singh et al. has demonstrated the use of Pt ALD as a means of modifying cobalt catalysts supported on silica gel for the production of higher alcohols (J.A. Singh, N. Yang, X. Liu, C. Tsai, K. Stone, B. Johnson, A.L. Koh & S.F. Bent, unpublished information). The controlled addition of Pt nanoparticles through ALD in that example resulted in increased selectivity to the valuable oxygenate products such as methanol and ethanol during syngas conversion. Feng et al. (90) synthesized Pd nanoparticles using Pd ALD on ZnO-coated silica gel and Al₂O₃-coated silica gel and demonstrated the catalysts for methanol decomposition. Liang et al. (91) likewise synthesized Pd nanoparticle catalysts using Pd ALD on Al₂O₃. The resulting catalysts were found to be similar in activity to commercial catalysts for the liquid-phase oxidation of glucose to gluconic acid. Ir particles deposited by ALD

have been shown to be more active and selective for decalin ring opening than those prepared using IWI (53).

In addition to monometallic nanoparticle catalysts, bimetallic catalysts are also of great interest. Traditional catalyst synthesis can struggle in creating uniform bimetallic catalysts, with the resulting catalysts often consisting of bimetallic particles of varying composition mixed together with monometallic particles (66). In principle, ALD can be used to create uniform bimetallic nanoparticles in alloy and core-shell configurations (66). However, to achieve bimetallic catalyst nanoparticles using ALD, strategies must be employed to ensure that the deposition of each material is controlled to occur only at certain surfaces, i.e., is area selective. Once the first metal is deposited, there is a heterogeneous surface for growth of the second metal composed of the existing metal and the surrounding support (typically an oxide). In the case where the material being deposited has little preference for growth on either material, one creates a catalyst with a mixture of active sites where there are bimetallic sites as well as monometallic sites of one or both of the metals.

To overcome this difficulty, there are a few strategies available for selective ALD, including use of SAMs to control the growth process (83). Cao et al. (83) used octadecyltrichlorosilane SAMs to create voids on the surface as a means of generating bimetallic nanoparticles using ALD. On the SAM-coated substrate, Pd nucleated in defects in the SAM layer (83). When a subsequent coating using Pt ALD was performed, Pt nucleated only on the Pd nanoparticles, as all other sites were blocked by the SAM, resulting in area selective growth. A second, simpler strategy uses chemical selectivity of the deposition process. Careful choice of process conditions can allow for selective growth, forming bimetallic particles (66). In particular, choice of precursor chemistry and temperature plays a strong role. For example, in the case of Pd ALD, surface H^* species are the active sites, and for Pt, surface O^* species are the active sites (66). By controlling where these active sites are, the nucleation of each material can be controlled to ensure that each material grows only on the other, resulting in bimetallic nanoparticles. Weber et al. (79) used such an approach to achieve Pt/Pd bimetallic nanoparticles without the use of SAMs. With H_2 as a counter-reactant, Pd does not grow on Al_2O_3 ; however, Pd deposits on Pt surfaces because H_2 chemisorbs on Pt, allowing for this growth process (79). Hence, by performing ALD of Pt nanoparticles on Al_2O_3 followed by Pd ALD using the H_2 counter-reactant, bimetallic nanoparticles of Pt/Pd were selectively produced. Lu et al. (66) used a related approach to create bimetallic alloy and core-shell nanoparticles of PdPt, RuPt, and RuPd on Al_2O_3 . Similarly, Wang et al. (92) prepared core-shell Au@Pd nanoparticles supported on SiO_2 using area-selective ALD. The resulting catalysts were demonstrated to have activity for the oxidation of benzyl alcohol, whereas Pd and Au catalysts were individually inactive (92).

Modification of the Catalyst Support

Besides simply serving as a scaffold, catalyst supports can influence catalytic activity and selectivity in a variety of ways, such as by creating new active sites through formation of an interacting perimeter; changing surface acidity/basicity; modifying the electronic structure of the active phase; and, for metal nanoparticle catalysts, changing their size and morphology, or even encapsulating the metal nanoparticles through strong metal support interactions (93–96). ALD can be used to deposit a thin oxide film as a modification layer with fine control over thickness, composition, and termination, without undesirably affecting the porosity and surface area of the support.

Owing to these qualities, the modification of catalyst supports by ALD has enabled fundamental study of the role of the support materials as well as identification and creation of active sites. In one

example, thin Al_2O_3 and ZnO ALD layers were deposited on a SiO_2 support, followed by Pd ALD, for study as methanol decomposition catalysts (90). Despite a higher amount of Pd deposited, the activity of the ZnO -supported Pd catalyst was lower than that of the Al_2O_3 -supported catalyst and suffered from quick deactivation. Dissolution of Pd into the ZnO layer was suggested to account for the deactivation. Sereda et al. (97) investigated the effect of the thickness of an additional ALD- Al_2O_3 or ALD- TiO_2 support layer in vanadium oxide catalysts for oxidative dehydrogenation of propane. Vanadium oxide was deposited onto pristine silica, ALD-modified silica, and bulk Al_2O_3 and TiO_2 supports. The ALD-modified supports showed different effects in activity and selectivity for oxidative dehydrogenation of propane, compared with silica or bulk Al_2O_3 and TiO_2 . A single cycle of Al_2O_3 ALD was shown to suppress the temperature programmed reduction (TPR) pattern of V_2O_5 , increase propane oxidation activation energy, and improve propene selectivity. The addition of a second Al_2O_3 layer increased propane conversion but shifted selectivity toward combustion. More than two cycles of Al_2O_3 further changed neither activity nor selectivity. One cycle of TiO_2 ALD achieved similar propene selectivity as bulk TiO_2 with even higher conversion. Increasing the number of TiO_2 ALD cycles further increased propane conversion, but propene selectivity was reduced. This study demonstrates that even single cycles of ALD can alter the interaction between the active catalyst species and the support and thus change the catalytic behavior.

In another example, Lobo et al. (51) deposited ALD- CeO_2 and ALD- TiO_2 on an Al_2O_3 support prior to Pt deposition to prevent leaching of the alumina support material under high-temperature and -pressure aqueous-phase reforming of 1-propanol. The authors found that a TiO_2 and CeO_2 support layer led to higher Pt dispersion, although these additional support layers failed to prevent Pt nanoparticles from sintering. The activity was found to follow the sequence of $\text{Pt-TiO}_2 > \text{Pt-Al}_2\text{O}_3 > \text{Pt-CeO}_2$, which suggests that other effects besides Pt dispersion, e.g., electronic properties, may also play a role in determining catalytic activity.

Introduction of Promoters

ALD overcoat layers on supported metal catalysts can physically block certain surface sites and create new interface sites. Thus, overcoat layers can change catalytic activity and selectivity, improve thermal stability, and prevent leaching of metal nanoparticles. This section describes the application of ALD for overcoat layers to improve catalytic activity or selectivity and to elucidate reaction mechanisms. Stability enhancement is discussed in the following section.

For improving catalytic activity or selectivity, the overcoat layer essentially serves as a promoter. A promoter is a relatively small quantity of one or more substances that, when added to a catalyst, improves the activity, selectivity, or useful lifetime of the catalyst. The surface composition of a catalyst is critical in determining its performance (98). By using ALD to directly deposit the promoter on top of existing active nanoparticles, surface composition can be carefully controlled.

There are several examples in the literature in which ALD overcoatings were used to promote catalytic activity. In one early example, ALD- TiO_2 was deposited on the surface of Ni powder (99), and the TiO_2 nanoislands that formed on the Ni surface were found to increase the CO_2 reforming of CH_4 and remove carbon deposits. Moreover, using ALD to overcoat more than one oxide has the potential to combine the functions of different oxides to improve both stability and catalytic activity. Using this strategy, Alba-Rubio et al. (100) combined Al_2O_3 and NbO_x overcoatings on copper catalysts. An ALD- Al_2O_3 overcoat on a SiO_2 -supported Cu catalyst protected the base metal against sintering and leaching during hydrogenation of furfural in liquid phase. Calcination at 937 K after Al_2O_3 overcoating regenerated porosity and therefore catalytic activity. The authors then showed that additional ALD- NbO_x deposition on the $\text{Al}_2\text{O}_3/\text{Cu}/\text{SiO}_2$ catalyst maintained

the amount of exposed Cu and introduced new acid sites. The production of furfuryl butyl ether was enhanced by introducing this second functionality onto the existing Al_2O_3 overcoat.

Although the majority of promoters deposited by ALD have been metal oxides, nonmetallic promoters have also been demonstrated. Stempel et al. (101) investigated the role of phosphorus in vanadyl pyrophosphate catalysts for selective oxidation of n-butane to maleic anhydride. ALD-P was deposited using tris(dimethylamino)phosphine ($[\text{Me}_2\text{N}]_3\text{P}$) and ozone. Partial coverage of the V_2O_5 surface by ALD-P was found to suppress total oxidation and improve selectivity to maleic anhydride. Although no new crystalline phases were formed after ALD-P deposition, the authors proposed that phosphorus on the V_2O_5 surface may moderate the activity of V^{5+} sites and improve selectivity. Phosphorus deposited by ALD exhibited stronger chemical bonding with V_2O_5 , compared with phosphine and phosphate $\text{P}/\text{V}_2\text{O}_5$ formed by IWI, as a loss of phosphorus was observed for the impregnated catalyst but not for the ALD catalyst.

The activity and selectivity of supported metal catalysts can depend not only on the total concentration of active surface sites but also on the ratio between different types of surface sites. Therefore, one way to tune the activity and selectivity is by selectively blocking certain surface sites. The reaction between organometallic precursors and the substrate surface during ALD usually takes place by attacking surface hydroxyl groups or highly energetic sites, i.e., undercoordinated and defect sites. Feng et al. (37) found that ALD- Al_2O_3 preferentially grows on the edge and corner sites of Pd nanoparticles, leaving the Pd (111) sites accessible and catalytically active. In another study (102), such preferential ALD- Al_2O_3 growth on Pd step sites was used to improve furan selectivity from furfural hydrogenation and decrease furfural alcohol formation.

In addition to blocking certain surface sites, the ALD overcoat layer can also influence the reduction and chemisorption properties of the underlying metal particle, which in turn changes the catalytic properties. Zhang et al. (103) used ALD- Al_2O_3 overcoat layers of different thicknesses on a copper chromite catalyst to achieve such a result. In that work, an Al_2O_3 overcoat considerably increased the reduction temperature of Cu^{2+} to Cu^0 , as evidenced by a broadening of the TPR features and a $\sim 300\text{-K}$ increase in the temperature required for complete reduction. The increased reduction difficulty may result from reduced diffusion owing to the overcoat layers or from copper aluminate formation. These catalysts were used in furfural hydrogenation, and the authors found that the activation energy decreased with increasing Al_2O_3 thickness. The proposed active species for furfural hydrogenation, based on operando X-ray absorption fine structure, is a combination of Cu^{1+} and Cu^0 . Ten cycles of Al_2O_3 ALD led to the highest Cu^{1+} concentration and increased furfural hydrogenation activity. The ALD overcoat layer also improved stability, possibly by minimizing coke formation owing to the microporous properties as well as retarding chromite migration by the presence of Al_2O_3 or copper aluminate formation.

Besides the promotion effects described above (introduction of new active sites, blocking specific surface sites, and influence on metal oxidation or reduction), catalytic activity and selectivity can also be tuned by changing the catalyst configuration, i.e., in what position the promoter is added. A comparison of different configurations can also shine light on the reaction mechanism and guide rational design to optimize formation of the desirable product. Yang et al. applied MnO ALD on SiO_2 -supported Rh catalysts for syngas conversion to higher oxygenates, with MnO acting as a promoter (N. Yang, J.S. Yoo, J. Schumann, P. Bothra, J.A. Singh, E. Valle, J.K. Nørskov & S.F. Bent, unpublished information). The highest activity and total production of higher oxygenates were achieved with the ultrathin MnO deposited as a support layer (i.e., before the Rh). The effect was attributed to formation of Rh-MnO interface sites that facilitated CO dissociation and stabilized transition states for higher oxygenate formation. In contrast, MnO deposited as an overcoat layer on Rh only moderately improved activity and selectivity, with the weaker effect attributed to the instability of MnO on the Rh surface upon CO adsorption.

In another example, a ZnO-promoted Pt/Al₂O₃ catalyst was synthesized by ALD for aqueous-phase reforming of 1-propanol (86). ZnO was deposited by ALD either as a support layer on Al₂O₃ prior to Pt ALD or as an overcoat layer after Pt ALD. The highest activity and H₂ selectivity were achieved on the ZnO-overcoated samples (ZnO/Pt/Al₂O₃), as they benefited from having both Pt-ZnO and Pt-Al₂O₃ interfaces. The other configuration, with ZnO as the support layer (Pt/ZnO/Al₂O₃), showed higher H₂ selectivity than the unpromoted Pt/Al₂O₃, but the activity was lower. The authors suggested that the overcoating ZnO layer generates steric hindrance on the surface and hence limits C-C bond activation, leading to the highest propanal selectivity. Both ZnO/Pt/Al₂O₃ and Pt/ZnO/Al₂O₃ showed improved thermal stability, which was attributed to the smaller lattice mismatch between Pt and ZnO.

Improving Stability

One of the major considerations of catalyst design is stability. The active metal nanoparticles in heterogeneous catalysts often suffer from deactivation owing to chemical, thermal, or mechanical causes (104). Such deactivation increases manufacturing cost from catalyst replacement and process shutdown and also impedes the industrialization of novel catalysts. Many industrial practices exist for alleviating deactivation, including optimizing the reaction process, choosing appropriate support materials, adding promoters, optimizing the pore structure, and improving the preparation of the active phase (104). Encapsulating nanoparticles can also improve the stability of metal nanoparticles against sintering and leaching (105, 106). Compared with other encapsulation methods, ALD can be used to synthesize protection layers with improved thickness control of the overcoat layer and modified surface properties to prevent coking and sintering simultaneously (107). Overcoat layers can be either deposited on the entire catalyst (i.e., on both the metal nanoparticle and the support) or selectively deposited surrounding the metal nanoparticle to minimize blocking of active sites (108). Post-treatment may be required to generate pores inside the overcoat film and expose catalytically active sites (107).

The literature contains several examples of successful stability enhancement using ALD overcoatings. Lu et al. (107) applied an ~8-nm ALD Al₂O₃ overcoat to stabilize Pd/Al₂O₃ during oxidative dehydrogenation of ethane at high temperature. Calcination after the Al₂O₃ overcoat regenerated the original mesopores and created new, smaller pores in the Al₂O₃ on the catalyst surface, making the Pd surface accessible to reagents. The overcoat layer was found to effectively inhibit coke formation and sintering of Pd nanoparticles. The ethylene yield was increased by an order of magnitude on the overcoated catalyst. CO adsorption studies revealed that Al₂O₃ was preferentially deposited on undercoordinated Pd sites, which otherwise led to the formation of coke and undesired products. Furthermore, the authors proposed that an Al₂O₃ overcoat with small pore channels may also break up large Pd surface ensembles, limiting the formation of large carbon filaments and reducing the probability of bimolecular reactions that could lead to coking. In another study, ALD TiO₂ was deposited on nanoporous gold to stabilize its structure (109). The overcoat layer increased CO oxidation activity by threefold compared with the uncoated counterpart. Annealing at 973 K was found to be necessary to break the TiO₂ overcoat film into nanocrystals, which is likely to enhance O₂ dissociation and therefore increase CO oxidation activity.

Many liquid-phase reactions also rely on heterogeneous catalysts, including various reactions in specialty chemical production (110) and biomass conversion (111). In addition to sintering, catalysts used in liquid-phase reactions can suffer from leaching of the active metal components and the support. Such irreversible deactivation causes detrimental environmental effects (112, 113) and also impedes the use of cheap and abundant base metals (e.g., Co, Cu) (114). ALD has gained

increasing interest in improving the stability of liquid-phase heterogeneous catalysts. O'Neill et al. (114) first reported the application of an ALD overcoat layer to stabilize base metal catalysts during liquid-phase reaction conditions. In their study, 45 cycles of ALD- Al_2O_3 were deposited on $\text{Cu}/\text{Al}_2\text{O}_3$, and the overcoat prevented sintering and leaching of the underlying Cu nanoparticles during furfural hydrogenation. Calcination at 973 K facilitated the crystallization of the Al_2O_3 overcoat layer and exposed active copper sites. Based on infrared spectroscopy characterization of CO adsorption on the $\text{Al}_2\text{O}_3/\text{Cu}/\text{Al}_2\text{O}_3$ surface, the authors determined that the ALD- Al_2O_3 preferentially covered the undercoordinated Cu sites, the sites that mainly contribute to sintering and leaching. An operando X-ray absorption spectroscopy study (115) showed that sintering was the major deactivation mechanism in uncoated catalysts and that the Al_2O_3 overcoat was able to maintain the Cu nanoparticle size during the reaction. However, the study also revealed that the Al_2O_3 overcoat introduced acidic sites and led to reversible deactivation of the catalyst from carbon deposition and pore blocking. In a follow-up study (116), the authors decreased the number of ALD cycles from 45 to 5 and achieved higher activity while maintaining stability. Thinner ALD films also lowered the calcination temperature required to expose the copper sites.

In a separate study, an ALD- TiO_2 overcoat was applied to stabilize Co/TiO_2 in aqueous-phase hydrogenation of biomass feedstock (117). The TiO_2 overcoat layer underwent densification and formed large pores (~ 16 nm) on the Co surface after calcination so that the activity of the $\text{TiO}_2/\text{Co}/\text{TiO}_2$ catalyst even increased after regeneration, owing to the exposure of more Co active sites. Both leaching and sintering were prevented by the thin TiO_2 overcoat layer.

Besides inert oxides, more reactive catalytic layers have also been deposited as overcoatings with the aim of improving stability and creating new active sites. Onn et al. (118) reported that ~ 1 -nm-thick ZrO_2 deposited by ALD on a $\text{Pd}/\text{Al}_2\text{O}_3$ catalyst stabilized the catalyst against sintering at high temperature. High-temperature calcination led to crystallization of the ZrO_2 overcoat layer and a semicore-shell structure. The methane oxidation activity of the $\text{ZrO}_2/\text{Pd}/\text{Al}_2\text{O}_3$ catalyst increased with increasing calcination temperature, whereas the activity of unmodified $\text{Pd}/\text{Al}_2\text{O}_3$ showed the opposite trend. Besides physically covering the Pd surface to stabilize the nanoparticles, the authors hypothesized that the proximity between ZrO_2 and Pd may also stabilize the PdO phase, which has to be present in the active phase for methane oxidation to proceed.

Although Al_2O_3 has been most commonly used as the overcoat layer, and many successful cases have been reported in which catalyst stability was enhanced, several factors—including the composition of the overcoat material, its thickness, and the calcination temperature—should be carefully considered to avoid the formation of undesired phases or adverse changes in surface properties. For example, in the work by O'Neill et al. (116), Al_2O_3 overcoat layers successfully stabilized Cu nanoparticles under furfural hydrogenation, whereas MgO_x failed to improve stability. However, Al_2O_3 introduced acidic surface sites and resulted in resinification and pore blocking (115). The authors showed that additional deposition of MgO_x on $\text{Al}_2\text{O}_3/\text{Cu}/\text{Al}_2\text{O}_3$ decreased surface acidity and retarded deactivation. Another study reported that Co and an Al_2O_3 overcoat can form cobalt aluminate after high-temperature calcination, so the resulting Co catalyst no longer shows activity for aqueous-phase hydrogenation of biomass feedstock (117). An ALD- SiO_2 overcoat on Au/TiO_2 also inhibited Au nanoparticles from sintering in CO oxidation (119). However, SiO_2 blocks active Au surface sites, and the resulting catalyst was not particularly active for CO oxidation, although heat treatment in O_2 -He at 300°C was able to increase catalytic activity, likely owing to removal of organic moieties from the ALD precursor. The authors of that study also compared SiO_2 overcoat layers deposited by ALD with those prepared by solution-phase grafting. Solution-based synthesis stabilized the Au nanoparticles and also maintained catalytic activity. The reduced activity from the ALD overcoat layer probably results from the dense SiO_2 film adversely passivating active Au sites.

Three-Dimensional Nanoreactors and Other Next-Generation Designs

There is growing interest in developing next-generation catalysts that can provide greater control over activity and selectivity. Conventional catalysts often follow scaling relations in which new compositions among a group of catalysts, e.g., metal alloys, follow a general trend based upon surface descriptors that limits both activity and selectivity in a reaction (120). New systems, such as ceramics, nitrides, and other less-explored materials, as well as unconventional designs, are one way to shift to more favorable scaling relations and thus higher selectivities and activities than those afforded using traditional catalysts (120). Whereas conventional catalysts typically consist of an active phase (e.g., a metal nanoparticle) on a support (typically a metal oxide), next-generation catalysts are expected to control the local environment around the catalytic site. Catalysts with an engineered local geometry may follow more favorable scaling relations, as the intermediates can be stabilized or destabilized based on the surrounding environment (121). This geometry surrounding the active nanoparticles or sites can take many forms, from simple size-confinement effects (122) to multiple active sites (123), and may appear as complex 3D structures (123), pores (122), and channels (121).

With the capability to tailor both the active catalytic site and its surrounding environment, ALD allows for the synthesis of next-generation catalysts. In some designs, one desires to deposit a specific environment around a catalytic site without completely coating the underlying catalytic site. By using area-selective ALD, such structures can be achieved, and some early examples already show promise. Ligand poisoning and SAMs are two schemes that have been used to protect a metal site from being fully coated during ALD (124). Lu et al. (67) demonstrated this methodology with Pd nanoparticles on Al_2O_3 and TiO_2 . After each $\text{Pd}(\text{hfac})_2$ exposure during Pd ALD, the growing Pd particles have hfac ligands on their surface, which prevents growth of Al_2O_3 and TiO_2 using exposures of trimethylaluminum and titanium isopropoxide, respectively. Because growth is inhibited on the hfac-coated Pd surface, deposition occurs only on the support. Other work has shown that ALD has the potential for creating novel engineered catalytic sites with both a controlled local chemistry and morphology. Work by Canlas et al. (122) created a shape-selective catalyst by using ultrathin coatings of ALD Al_2O_3 . The process is outlined in **Figure 3**. In this scheme, a template molecule was graphed onto the surface to define the active site (122). A thin metal oxide coating was then deposited using ALD. Finally, ozone removed the initial organic template molecule. TiO_2 catalysts with patterned ALD Al_2O_3 coatings prepared using this technique had a higher selectivity for the oxidation of benzyl alcohol versus 2,4,6-trimethylbenzyl alcohol (122).

In addition to controlling the geometry, enhanced performance can be obtained by adding catalytic sites to activate different reactants. Increased activity was observed for the hydrogenation of nitrobenzene with hydrogen on nanotube catalysts prepared using ALD. The catalysts consisted of Pt/TiO_2 as the outer wall and $\text{NiO}/\text{Al}_2\text{O}_3$ as the inner wall with all layers prepared using ALD (123). The local confinement allows activated hydrogen from hydrazine decomposition on Ni-based sites to spill over to Pt sites where nitrobenzene is hydrogenated. These ideas can be further extended to have multiple ALD layers that are chosen with different chemical functionalities to interact with the different functional groups of the reactants, products, and intermediates.

Another next-generation catalyst design involves exerting control over which crystal faces of the catalyst are exposed. There are orders-of-magnitude differences in activity between different crystal facets (125–127). Traditional wet chemical synthesis can create beautifully complex geometries, such as cubes that have different ratios among the various exposed crystal planes (128–130); however, such techniques may be complex and costly and cannot be directly translated between different materials. ALD coatings provide an alternative method for controlling which crystal

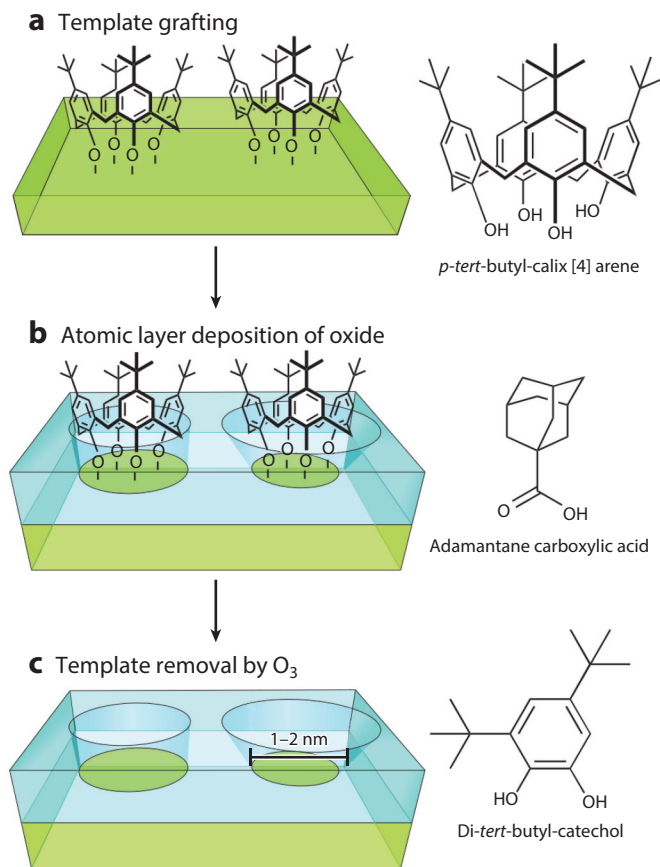


Figure 3

Scheme for the synthesis of shape-selective catalysts by atomic layer deposition (ALD). Initially, a template organic molecule is deposited. ALD is performed around the molecule and the template molecule is removed using ozone. The catalyst will react only with molecules small enough to fit within the resulting pores. Reprinted by permission from Macmillan Publishers Ltd., *Nature Chemistry* (122), ©2012.

planes of a catalyst are exposed by selectively depositing material on some crystallographic facets but not others.

Theoretical and experimental work demonstrates this emerging application of ALD. Lu et al. (131) performed density functional theory calculations to study the preference of Al₂O₃ ALD between Pd (111) terraces and stepped Pd (211) surfaces. On the Pd (211) surface, the reaction was found to be more energetically favored and the *CH₃ group to bind less strongly, resulting in a preference for growth on the (211) over the (111) surface. Using similar ideas, one could perform ALD to block certain crystal planes and obtain catalysts with the only crystal planes exposed being those with the desired selectivity. Ding et al. (132) demonstrated this concept using Ga₂O₃ ALD to decorate Pd nanoparticles supported on γ-Al₂O₃. ALD-synthesized Pd nanoparticles were coated using Ga₂O₃ ALD using Ga(CH₃)₃ and O₂ for application in the catalytic hydrogenation of acetylene to ethene. Ga₂O₃ was shown to selectively coat the edge and kink sites owing to its preference for growth on high-energy Pd sites, while leaving the Pd (111) surfaces exposed. By increasing the fraction of Pd (111) sites, the resulting catalysts had both high activity and selectivity

for the reaction (132). This scheme is not expected to be limited to Pd, as such low-coordination sites are ubiquitous on nanoparticles. By controlling exposed crystal faces using ALD, the active catalytic surface can be finely tailored.

ALD also provides a means for creating subnanometer catalysts with narrow particle size distributions. Extremely small metal nanoparticles and single atom sites offer opportunities for higher performance and unique reactivity (133). As the catalyst particles decrease in size, the same active surface area can be achieved with much lower usage of metals, which is especially important in the case of precious metals (133). Moreover, with decreasing particle size, catalyst behavior strongly changes owing to increased fractions of undercoordinated metal atoms, quantum size effects, and the high percentage of atoms in close proximity to the support (133). Conventional wet synthetic methods for subnanometer particles require post-treatment to remove ligands or protective encapsulation. The possible side effects are nontrivial for producing catalysts with such small particles, owing to the risk of restructuring or sintering (133). Other techniques, such as mass-selected soft landing, have significant limitations in expense, throughput, or yield (67, 133) or are not appropriate for coating high-surface-area supports typically used in heterogeneous catalysis (67). In contrast, ALD does not typically require post-treatments as in the case of wet chemistry, and unlike mass-selected soft landing, it does not require ultrahigh vacuum and allows for coating typical catalyst supports. Using these favorable properties, recent studies have demonstrated ALD as a means to make highly dispersed, subnanometer, catalytic nanoparticles (80, 87, 134). Sun et al. (87) prepared catalysts on graphene by depositing single atoms and subnanometer clusters of Pt using ALD. The study demonstrated ALD's ability to control size distributions even in the case of extremely small particle sizes by changing the number of metal ALD cycles (87). The small size resulted in an order-of-magnitude increase in activity relative to a commercial Pt/C catalyst for electrochemical methanol oxidation owing to low-coordination sites and unoccupied metal d orbitals on the prepared catalysts (87).

OPPORTUNITIES AND CHALLENGES

ALD has begun to show great potential in advancing the design, synthesis, and understanding of heterogeneous catalysts for a variety of reactions, including those in both gas and liquid phase. ALD is capable of depositing nanoparticles and thin films with excellent control over size, composition, and thickness, all properties that are important for catalysts. By applying ALD to different components of catalytic systems—including the synthesis of nanoparticles or thin-film catalysts, the modification of the catalyst support, and the deposition of overcoat layers—improved catalytic activity, selectivity, and stability have been achieved. Several examples were highlighted in this review.

With its atomic-level control over catalyst composition and structure, ALD is poised to contribute to an improved fundamental understanding of important catalysts as well as aid in the rational design of the desirable catalytic sites. By providing catalysts with tunable material composition and a high level of uniformity, ALD may facilitate the identification of active sites and the evaluation of structure-property relationships. The capability of ALD for uniform deposition on both planar and powder substrates can help bridge the gap between model catalysts and practical catalysts. Furthermore, the recently reported integration of an ALD reactor with reaction testing (135) is a promising new direction toward allowing catalyst synthesis and performance analysis without exposing the catalyst to air and moisture, which could improve understanding of well-defined catalysts under working conditions.

Despite the unique capabilities of ALD, other factors must be considered when applying ALD to heterogeneous catalysis. Most ALD-synthesized or -modified catalysts can be readily used in

catalytic reactions, but additional treatment may be necessary to allow access to the active catalyst surface (100, 118), to create strong interaction between metal and oxide, and to achieve the desired crystal structure or oxidation state. ALD processes are typically conducted in a vacuum at elevated temperatures, and to avoid undesirable structural changes of the catalyst substrate as well as degradation or poisoning from residual ligands, the choice of ALD precursors, counter-reactants, and growth conditions may need to be optimized depending on the specific catalysis systems. It is also worth noting that in some reported cases, ALD was inferior to IWI. For example, Cronauer et al. (136, 137) studied Co catalysts for Fischer-Tropsch synthesis, where Co nanoparticles as well as Pt, Ru, and Ir promoters were synthesized by both ALD and IWI. The ALD catalysts showed lower activity than the IWI catalysts. The authors attributed this inferior activity to the larger Co size in the ALD-grown catalyst and lower-than-expected noble metal loading deposited by ALD.

Regarding its industrial application, the cost and throughput of ALD are considerations given the nature of the process. Effective deposition on powders at large scale or of nanometer size could require specialized ALD equipment, such as a rotary reactor (44), fluidized bed reactor (43), or spatial ALD reactor (45). In addition, the cost of commonly used ALD precursors is often higher than compounds used for impregnation or colloidal synthesis. Although these factors may hinder the commercial application of ALD in heterogeneous catalysis, if ALD can better target the desired catalyst composition and structure, it may provide outstanding catalyst properties that are not afforded by other synthesis methods. Moreover, in many cases, a low number of ALD cycles is effective in improving catalytic performance; thus, the cost of materials and time may not be forbidding. The expense of ALD equipment and precursors could be driven down by the rapidly increasing application of ALD in areas such as microelectronics, photovoltaics, batteries, and fuel cells. However, notwithstanding eventual possibilities for commercialization, ALD-synthesized catalyst systems promise extraordinary potential for fundamental studies in which to gain atomic-scale understanding and guidance for rational catalyst design in a variety of important conversion systems.

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